

Part A: MCQ Section
Part B: Hands-On Programming Section

Prepare for the Amazon ML Summer School selection test with these practice questions, curated from 2024 test feedback. Star the repository to support our efforts! Happy learning! 😊

Part A: Multiple Choice Questions (Machine Learning and Statistics)

Q1. The following confusion matrix describes the performance of a classification model predicting diabetes status for a dataset with $n = 192$.

	Predicted: 0	Predicted: 1
Actual: 0	TN = 118	FP = 12
Actual: 1	FN = 47	TP = 15

What is the estimated **False Positive Rate (FPR)** for this confusion matrix?

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

Options:

- A. 0.0923
- B. 0.0045
- C. 0.0872
- D. 0.1200

Q2. A Machine Learning specialist is tasked with classifying a large dataset of cosmetic products into various categories based on similarities, without prior information on product names or types. Which algorithm is most suitable for this task?

Options:

- A. Decision Tree Model
- B. Logistic Regression Algorithm
- C. K-Means Clustering
- D. Linear Regression

Q3. A bag contains 8 red balls and 6 black balls. If two balls are drawn one after the other **without replacement**, what is the probability that the second ball is red, given that the first ball was black?

A. $\frac{8}{13}$

B. $\frac{25}{13}$

Options:

C. $\frac{7}{13}$

D. $\frac{11}{13}$

Q4. Consider a simple linear regression model representing the relationship between study hours (x) and exam scores (y), with a slope of 5 and an intercept of 10. What is the equation of the regression line?

Options:

- A. $y = 5x + 10$
- B. $y = 10x + 5$
- C. $y = 105 + b$
- D. $y = 5 - 10b$

Q5. Given the matrix:

$$\begin{bmatrix} 1 & x - 1 \\ 3 & 2x + 3 \end{bmatrix}$$

Which of the following values of x makes the matrix **non-invertible**?

Hint: A matrix is non-invertible if its determinant is zero.

Options:

- A. 2
- B. 6
- C. 8
- D. 9

Q6. Evaluate the integral:

$$\int \frac{2}{\sin x + \sqrt{3} \cos x} dx$$

Options:

- A. $\log \left| \csc \left(x + \frac{\pi}{3} \right) - \cot \left(x - \frac{\pi}{3} \right) \right| + C$
- B. $\log \left| \csc \left(x - \frac{\pi}{3} \right) - \cot \left(x + \frac{\pi}{3} \right) \right| + C$
- C. $\log \left| \csc \left(x + \frac{\pi}{3} \right) - \cot \left(x + \frac{\pi}{3} \right) \right| + C$
- D. $\log \left| \csc \left(x - \frac{\pi}{3} \right) + \cot \left(x - \frac{\pi}{3} \right) \right| + C$

Q7. The frequency distribution of the number of letters in last names from a Name Registry Library is shown below:

Alphabet Count	Frequency
1-4	5
4-7	10
7-10	50
10-13	25
13-16	5
16-19	5

Calculate the **mean** number of letters in the last names.

Options:

- A. 10
- B. 6.25
- C. 8
- D. 9.45

Q8. James and Simon are solving a math problem. The probability that James solves it is $\frac{3}{4}$, and the probability that Simon solves it is $\frac{4}{5}$. What is the probability that **neither** solves the problem?

Options:

- A. $\frac{1}{20}$
- B. $\frac{3}{20}$
- C. $\frac{13}{20}$
- D. $\frac{2}{15}$

Q9. An equal number of chocolates are packed into packets for gifts. After packing, 10 chocolates remain. If the number of chocolates per packet is increased by 2, there is a shortage of 4 chocolates. How many packets are there?

Options:

- A. 3
- B. 5
- C. 7
- D. 9

Q10. John, Peter, and Tom each have a one-rupee coin. If they toss their coins simultaneously, what is the probability that all coins show the **same face** (all heads or all tails)?

Options:

- A. $\frac{1}{4}$
- B. $\frac{1}{6}$
- C. $\frac{1}{8}$
- D. $\frac{1}{12}$

Q11. Given $f(x) = 1 + 2^x \log 2 \frac{d}{dx} G(x)$ and $G(2) = 8$, find the value of $G(3)$.

Options:

- A. 9
- B. 11
- C. 13
- D. 15

Q12. The weights (in kilograms) of 10 students in a class are given below:

Student	Weight (Kg)
A	45
B	50
C	52
D	55
E	60
F	62
G	65
H	68
I	70
J	75

Calculate the **mean weight** of the students.

Options:

- A. 58.5
- B. 60.2
- C. 62.6
- D. 64.6

Part B: Hands-On Programming (Data Structures and Algorithms)

Problem 1: Sum of Array

Problem Statement Peter has an integer array A of size N . He performs the following operations:

1. Update each element $A[i]$ with the **GCD** of $A[i]$ and the **largest element** in the subarray $A[0..i]$.
2. Group the elements such that the **largest element** is paired with the **smallest element**, the **second largest** with the **second smallest**, and so on.
3. Compute the **GCD** of each group and store the results in a new array B .
4. Return the **sum** of array B .

Note:

- N is always even.
- The index i ranges from 0 to $N - 1$.

Input Specification

- input1: Integer N , the size of array A .
- input2: Integer array A .

Output Specification

- Integer representing the sum of array B .

Example

```

Input: N = 4, A = [12, 15, 3, 10]
# Step 1: Largest in subarrays: [12], [15], [15], [15]
# GCD updates: [GCD(12,12)=12, GCD(15,15)=15, GCD(3,15)=3, GCD(10,15)=5]
#           -> A = [12, 15, 3, 5]
# Step 2: Pair largest (15) with smallest (3), second largest (12) with second smallest (5)
# Step 3: GCD(15,3)=3, GCD(12,5)=1
# Step 4: Sum of B = [3, 1] = 4
Output: 4

```

Problem 2: Minimum Number of Key Presses

Problem Statement George uses a special keyboard with 11 numeric keys: $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 00\}$.

The monitor initially displays 0.

- Pressing 00 multiplies the current value by 100.
- Pressing any other key multiplies the current value by 10 and adds the key's value.

Given a numeric string S , find the **minimum number of key presses** to display the number represented by S .

Input Specification

- input1: String S , representing the target number.

Output Specification

- Integer representing the minimum number of key presses.

Example 1

```

Input: "60004"
# Steps:
# 0 -> Press 6:  $(0*10)+6 = 6$ 
# 6 -> Press 00:  $6*100 = 600$ 
# 600 -> Press 0:  $(600*10)+0 = 6000$ 
# 6000 -> Press 4:  $(6000*10)+4 = 60004$ 
Output: 4

```

Example 2

```

Input: "1355506027"
# Steps:
# 0 -> Press 1:  $(0*10)+1 = 1$ 
# 1 -> Press 3:  $(1*10)+3 = 13$ 
# 13 -> Press 5:  $(13*10)+5 = 135$ 
# 135 -> Press 5:  $(135*10)+5 = 1355$ 
# 1355 -> Press 5:  $(1355*10)+5 = 13555$ 
# 13555 -> Press 0:  $(13555*10)+0 = 135550$ 
# 135550 -> Press 6:  $(135550*10)+6 = 1355506$ 
# 1355506 -> Press 0:  $(1355506*10)+0 = 13555060$ 
# 13555060 -> Press 2:  $(13555060*10)+2 = 135550602$ 
# 135550602 -> Press 7:  $(135550602*10)+7 = 1355506027$ 
Output: 10

```

Problem 3: Sam's Dilemma

Problem Statement Given three integer arrays A , B , and C , create all possible triplets by selecting one

integer from each array. For each triplet, compute the **XOR** of the three numbers and count how many triplets have an **even number of set bits** in the XOR result.

Note: Set bits are the number of 1s in the binary representation of a number.

Input Specification

- input1: Integer array A .
- input2: Integer, length of array A .
- input3: Integer array B .
- input4: Integer, length of array B .
- input5: Integer array C .
- input6: Integer, length of array C .

Output Specification

- Integer representing the number of triplets with an even number of set bits.

Example

```
Input: A = [1, 2], len(A) = 2, B = [3], len(B) = 1, C = [4, 5], len(C) = 2
# Triplets: (1,3,4), (1,3,5), (2,3,4), (2,3,5)
# XOR results: 6 (110, 2 set bits), 7 (111, 3 set bits), 5 (101, 3 set bits), 4 (100, 1 set bit)
# Even set bits: Only 6 has 2 set bits
Output: 1
```

Problem 4: The Last Chocolate

Problem Statement Bruce plays a game with Andrew using N chocolates in a single-row box and an

unbiased coin. For each toss:

- **Heads:** Remove the **first** chocolate.
- **Tails:** Remove the **last** chocolate.

Given an integer K , find the number of possible coin-toss sequences that leave the K -th chocolate as the last remaining after $N - 1$ tosses.

Note: Use 1-based indexing.

Input Specification

- input1: Integer N , number of chocolates.
- input2: Integer K , position of the last chocolate.

Output Specification

- Integer representing the number of valid toss sequences.

Example

```
Input: N = 3, K = 2
# Chocolates: [1, 2, 3]
# Sequences leaving 2 as last:
# 1. Heads, Tails: Remove 1, then 3 -> [2] remains
# 2. Tails, Heads: Remove 3, then 1 -> [2] remains
Output: 2
```

Additional Notes

Some questions in the 2024 selection test included image-based content (e.g., charts, graphs). These are not included here but can be found in the repository. Practice thoroughly and explore additional resources to prepare for the 2025 test. Good luck! 😊